STUDYHUB

CLASS 10 — MP BOARD

SUBJECT: SCIENCE

SAMPLE QUESTION PAPER — SET 2

Time: 3 Hours	Maximum Marks: 75
Medium: English	Set: 2

Note: This is an original StudyHub practice paper based on the MPBSE syllabus and examination pattern. It is not an official MPBSE paper.

GENERAL INSTRUCTIONS

1. All questions are compulsory.
2. Questions 1 to 5 are objective-type questions. Each question carries 6 marks.
3. Internal choices are given from Questions 6 to 23.
4. Answer Questions 6 to 17 in about 30 words each.
5. Answer Questions 18 to 20 in about 75 words each.
6. Answer Questions 21 to 23 in about 120 words each.
7. Draw neat and labelled diagrams wherever required.
8. Balance chemical equations wherever required.

PART — A

OBJECTIVE TYPE QUESTIONS

Q.1 Choose the correct alternative. [6 Marks]

(1) Which gas is evolved when a metal reacts with a dilute acid?

A. Oxygen
B. Hydrogen
C. Nitrogen
D. Carbon dioxide

(2) Which salt is commonly used in the manufacture of glass, soap and paper?

A. Sodium carbonate
B. Sodium chloride
C. Calcium sulphate
D. Potassium nitrate

(3) Which functional group is present in ethanol?

A. –COOH
B. –CHO
C. –OH
D. –CO

(4) Which part of the brain controls involuntary actions such as heartbeat and breathing?

A. Cerebrum
B. Cerebellum
C. Medulla
D. Spinal cord

(5) Which phenomenon is responsible for the splitting of white light into its constituent colours?

A. Reflection
B. Dispersion
C. Diffraction
D. Absorption

(6) The SI unit of electric current is—

A. Volt
B. Ohm
C. Ampere
D. Watt

Q.2 Fill in the blanks. [6 Marks]

(1) The process of coating iron with zinc to prevent rusting is called ________.

(2) The chemical formula of bleaching powder is ________.

(3) The conversion of sugar into alcohol by microorganisms is called ________.

(4) The male reproductive part of a flower is called the ________.

(5) The transparent front part of the human eye is called the ________.

(6) The device used to measure electric current in a circuit is called an ________.

Q.3 Write True or False. [6 Marks]

(1) Corrosion of iron requires oxygen and moisture.

(2) Sodium is stored under water because it reacts violently with air.

(3) Methane contains a double bond between carbon atoms.

(4) The pancreas secretes insulin.

(5) The image formed by a plane mirror is laterally inverted.

(6) A fuse wire should have a very high melting point.

Q.4 Match the following. [6 Marks]

Column A	Column B
(i) Plaster of Paris	(a) Hormone
(ii) Ethene	(b) $CaSO_4 \cdot \frac{1}{2}H_2O$
(iii) Adrenaline	(c) Unsaturated hydrocarbon
(iv) Stomata	(d) Gaseous exchange
(v) Fuse	(e) Safety device
(vi) Solenoid	(f) Magnetic field

Q.5 Answer in one word or one sentence. [6 Marks]

(1) Name the process by which a pure metal is obtained from its ore.

(2) What is the chemical formula of methane?

(3) Name the organ responsible for pumping blood throughout the human body.

(4) Which hormone regulates blood sugar level?

(5) What is the phenomenon of bending of light when it passes from one transparent medium to another called?

(6) Name one biodegradable waste material.

PART — B

SHORT ANSWER QUESTIONS

Q.6 What is a decomposition reaction? Give one example. [2 Marks]

OR

Why are chemical equations balanced? [2 Marks]

Q.7 Why does dry HCl gas not change the colour of dry blue litmus paper? [2 Marks]

OR

What is a universal indicator? [2 Marks]

Q.8 Why is potassium kept immersed in kerosene oil? [2 Marks]

OR

What is an alloy? Give one example. [2 Marks]

Q.9 What is a covalent bond? [2 Marks]

OR

Write the molecular formula and structural formula of ethene. [2 Marks]

Q.10 What is the role of bile juice in digestion? [2 Marks]

OR

Why is haemoglobin important for our body? [2 Marks]

Q.11 What is the function of the spinal cord in reflex actions? [2 Marks]

OR

Name any two plant hormones and write one function of each. [2 Marks]

Q.12 Differentiate between sexual and asexual reproduction on any two points. [2 Marks]

OR

What is vegetative propagation? Give one example. [2 Marks]

Q.13 What is a gene? [2 Marks]

OR

Why are acquired traits generally not inherited? [2 Marks]

Q.14 State the two laws of reflection of light. [2 Marks]

OR

What is the principal axis of a spherical mirror? [2 Marks]

Q.15 What is presbyopia? State one method of correcting it. [2 Marks]

OR

Why is the pupil of the eye important? [2 Marks]

Q.16 A wire carries a current of 3 A when connected to a potential difference of 18 V. Calculate its resistance. [2 Marks]

OR

Write any two factors on which the resistance of a conductor depends. [2 Marks]

Q.17 What is an electromagnet? Write one use of it. [2 Marks]

OR

State Fleming's Left-Hand Rule. [2 Marks]

PART — C

3-MARK QUESTIONS

Q.18 Explain the following terms with one suitable example each:
(i) Corrosion
(ii) Rancidity
(iii) Reduction [3 Marks]

OR

Explain the reactivity series of metals and state its importance in predicting displacement reactions. [3 Marks]

Q.19 Explain how water and minerals are transported from roots to different parts of a plant. [3 Marks]

OR

Explain the role of the small intestine in the absorption of digested food. [3 Marks]

Q.20 Explain why the Sun appears reddish during sunrise and sunset. [3 Marks]

OR

What is the power of a lens? Calculate the power of a lens whose focal length is 25 cm. [3 Marks]

PART — D

LONG ANSWER QUESTIONS

Q.21 Two resistors of 8 Ω and 12 Ω are connected in parallel across a 24 V battery.
(i) Calculate the equivalent resistance of the combination.
(ii) Calculate the total current drawn from the battery.
(iii) Find the current through the 8 Ω resistor. [4 Marks]

OR

An electric heater has a resistance of 20 Ω and is connected to a 220 V supply.
(i) Calculate the current flowing through the heater.
(ii) Calculate its electrical power.
(iii) Calculate the electrical energy consumed in 2 hours in kWh. [4 Marks]

Q.22 Draw a neat and labelled diagram of a human heart and explain the pathway of blood through its four chambers. [4 Marks]

OR

Explain the process of double circulation in human beings. Why is it advantageous for the body? [4 Marks]

Q.23 An object is placed 30 cm in front of a concave mirror having a focal length of 15 cm.
Using the mirror formula:
(i) Calculate the position of the image.
(ii) State the nature and size of the image.
(iii) Draw the appropriate ray diagram. [4 Marks]

OR

Explain the working principle of an electric motor. Draw a neat and labelled diagram and state the function of the split-ring commutator. [4 Marks]

ANSWER KEY

Q.1: (i) B (ii) A (iii) C (iv) C (v) B (vi) C

Q.2: (i) Galvanisation (ii) $CaOCl_2$ (iii) Fermentation (iv) Stamen (v) Cornea (vi) Ammeter

Q.3: (i) True (ii) False (iii) False (iv) True (v) True (vi) False

Q.4: (i)-(b), (ii)-(c), (iii)-(a), (iv)-(d), (v)-(e), (vi)-(f)

Q.5: (i) Metallurgy / extraction (ii) CH_4 (iii) Heart (iv) Insulin (v) Refraction (vi) Vegetable peels / paper

Q.16: 6 Ω.

Q.20 (OR): 4 D.

Q.21: 4.8 Ω; 5 A; 3 A.

Q.21 (OR): 11 A; 2420 W; 4.84 kWh.

Q.23: 30 cm in front; real, inverted and same size.

MARKS VERIFICATION

Section	Questions	Marks
Objective	1–5	30
Short Answer	6–17	24
3-Mark Questions	18–20	9
Long Answer	21–23	12
TOTAL	23 Questions	75

Final check: 30 + 24 + 9 + 12 = 75 Marks.